

ILRDVS: Complete Website Technology Stack & Engineering Reference

Comprehensive Technical Breakdown of Frontend, Backend API, Database, Multimodal AI Vision, Cryptographic Security, Geospatial Engine & Cloud DevOps

1. Architectural Overview & Design Philosophy

The **Intelligent Land Record Digitization and Validation System (ILRDVS)** is an enterprise digital governance solution built for the modern revenue ecosystem. The platform bridges the divide between archaic paper records (7/12 Satbara extracts, Ferfar mutation registers, Sale Deeds, Property Cards) and modern national digital infrastructure. Engineered with a **decoupled full-stack TypeScript architecture**, the platform enforces type safety across the entire pipeline: from client input forms to API transport payloads, database entity models, and vision extraction schemas.

2. Frontend Presentation Tier Architecture

The frontend is constructed using **Next.js 14** with the **React 18 App Router**. It provides responsive, accessible, and lightning-fast portals for three distinct user personas: **Citizens**, **Verifiers (Inspectors)**, and **Tehsildars (Revenue Officers / Admins)**.

- **Next.js 14 (v14.2.24) App Router**: Server-Side Rendering (SSR), Static Site Generation (SSG), and incremental static regeneration. Handles nested layout hierarchies (e.g. `app/portal/layout.tsx`), optimized image caching, and route-level error boundaries.
- **React 18 (v18.3.1)**: Concurrent React features including `useTransition`, `Suspense`, and automatic state batching. Powers high-responsiveness across complex form wizards and multi-pane inspection stations.
- **Tailwind CSS (v3.4.17) + Autoprefixer**: Utility-first design system customized with the official Government of India tricolor branding (Emerald Green #065f46, Deep Ashoka Blue #1e3a8a, Saffron #b45309). Glassmorphic panels with sub-pixel borders.
- **TanStack React Query (v5.64.2)**: Declarative server state management. Features automated background refetching, query key invalidation on mutation (e.g. invalidating `['verificationQueue']` on digital signature), optimistic updates, and offline caching.
- **Axios (v1.7.9)**: Centralized HTTP client configured with request interceptors for automatic JWT Bearer header injection and response interceptors standardizing backend error structures and cold-start wake-up notifications.
- **React Hook Form (v7.54.2) + @hookform/resolvers**: High-performance uncontrolled form state engine eliminating unnecessary re-renders. Integrates seamlessly with Zod resolvers for real-time validation across multi-step registration and document upload wizards.
- **i18next (v26.4.2) + react-i18next (v17.0.13)**: Full trilingual internationalization supporting Marathi (मराठी - primary state language), Hindi (हिन्दी), and English. Dynamic language toggle in top bar preserves user language preference across sessions.
- **QRCode (v1.5.4) & jsQR (v1.4.0)**: Two-way QR code engine. Generates high-density tamper-evident cryptographic verification QR codes on digital extracts; camera-based client-side scanner allows instant document authenticity audits.
- **JsBarcode (v3.12.3)**: Renders standard CODE128 and CODE39 statutory barcodes embedding Archival Serial Identifiers (e.g. ARCH-MH-REG-2026-1001) onto digitized records for physical file room tracking.
- **Recharts (v2.15.0)**: Declarative SVG charting library powering executive dashboards: land usage distribution donuts, monthly mutation throughput area charts, and verifier SLA turnaround bars.
- **Lucide React (v0.473.0)**: Consistent, lightweight SVG iconography tailored for enterprise governance interfaces.

3. Backend API Gateway & Microservices Architecture

The backend is built as a high-concurrency **Node.js (v22 LTS)** and **Express.js (v4.21.2)** RESTful application compiled via TypeScript 5.7. It encapsulates statutory land governance business logic, document streaming, OCR queues, and cryptographic signature validation.

- **Express.js (v4.21.2)**: Core web application framework. Modular routing structure (`routes/auth`, `routes/land-record`, `routes/verification`, `routes/document`), centralized error middleware, and sub-second endpoint execution.
- **TypeScript (v5.7.3) & tsx (v4.19.2)**: Full TypeScript strict mode enforcing interface contracts across request bodies, database entities, and internal services. `tsx` provides high-speed zero-config runtime execution for development and database seeding.
- **JSON Web Tokens (v9.0.2)**: Stateless, RFC 7519 compliant authentication. Signs user ID, role, and department into cryptographically secured tokens delivered via `HttpOnly`, `SameSite` cookies with fallback Bearer header authentication.
- **bcryptjs (v3.0.2)**: One-way adaptive cryptographic password hashing utilizing 10 salt rounds to resist rainbow table and brute-force attacks.
- **Multer (v1.4.5-lts.1)**: Streaming multipart/form-data middleware for document ingestion. Enforces MIME type verification (PDF, PNG, JPEG, TIFF) and strict 20MB file size quotas.
- **Sharp (v0.33.5)**: High-performance image processing engine based on libvips. Powers on-the-fly thumbnail generation, DPI normalization (300–400 DPI archival standard), contrast adjustment, and deskewing before OCR extraction.

- **pdf-parse (v1.1.1)**: Direct digital PDF stream parser extracting embedded vector text and metadata from born-digital government gazettes and e-Registration sale deeds.
- **Zod (v3.24.1)**: Dual-layer runtime validation. Validates request parameters, query strings, and multipart fields before controller invocation, automatically returning 400 Bad Request with field-level error messages.
- **CORS (v2.8.5)**: Dynamic origin validation allowing seamless cross-origin communication between the deployed Next.js frontend and Express backend across cloud domains (*.onrender.com, localhost).
- **Morgan (v1.10.0)**: Structured HTTP request logging recording method, route, response status, latency in milliseconds, and content length for operational telemetry.

4. Data Persistence Tier: PostgreSQL & Prisma ORM

The system utilizes **PostgreSQL 16** hosted on **Neon Serverless Cloud** (with Render Managed DB compatibility). Data access is managed through **Prisma ORM v6.4.1**, providing type-safe query building, connection pooling, and declarative schema migrations.

Core Relational Models:

- **User**: Central identity model supporting roles (ADMIN, OFFICER, VERIFIER, CITIZEN), account statuses (PENDING_VERIFICATION, PENDING_APPROVAL, ACTIVE), masked Aadhaar, and audit timestamps.
- **LandRecord**: Core parcel entity holding 14-digit ULPIN, survey/gat numbers, plot areas in hectares, tenure classification, ASR circle rate valuation, and multi-layer attributes.
- **Document**: Ingested archival scans with MIME types, storage paths, processing status (UPLOADED, PROCESSING, PROCESSED, NEEDS_REVIEW), OCR confidence scores, and JSONB extraction entities.
- **VerificationWorkflow & Record**: 4-Eye dual-control state machine capturing verifier remarks, tehsildar digital signature certificates, SHA-256 payload hashes, and QR verification tokens.
- **OfficerApplication**: Controlled officer onboarding workflows storing employee ID, departmental jurisdiction, and verification milestones.
- **AuditLog**: Append-only compliance ledger recording actor, action type, IP address, timestamp, and modified resources.

5. Hybrid AI Vision & Multilingual OCR Pipeline

ILRDVS deploys an intelligent **Hybrid AI Vision Pipeline** designed for both extreme accuracy and failover resiliency:

- **Cloud Primary: Google Gemini 1.5 Flash Vision API**: Evaluates high-resolution document scans using advanced multimodal reasoning. Capable of parsing complex handwritten Modi/Devanagari script, tabular Satbara extracts, and boundary demarcation text with sub-2-second turnaround time and $\geq 94\%$ confidence.
- **Edge Fallback: On-Device OpenCV + EasyOCR / Tesseract.js**: If cloud connectivity is disrupted or offline local processing is mandated for sensitive archival scans, the system automatically redirects documents to an on-device Python/Tesseract pipeline. Executes image deskewing, Otsu binarization, noise filtering, and local OCR without service downtime.
- **Automated Entity Normalization**: Extracts key values into typed structures: Vendor Name, Purchaser Name, Consideration Amount, Khata Number, Khasra Number, and Boundary Demarcations (East, West, North, South).

6. The 8-Layer Land Stack Geospatial Engine

Under DILRMP 3.0 guidelines, ILRDVS implements the unified **8-Layer Land Stack** anchored by the 14-digit Bhu-Aadhaar (ULPIN):

- **L1 Cadastral Map**: Geo-referenced WGS-84 vector polygons with sub-meter vertex pins.
- **L2 Record of Rights**: Authoritative legal title, tenure status, and ownership shares (7/12 & 8A).
- **L3 NGDRS Registration**: Registered conveyance deeds, e-Challan stamp duty numbers, and sale records.
- **L4 Land Use & Zoning**: Master plan zoning (Agricultural, Residential, Commercial, Industrial, CRZ buffer).
- **L5 NAKSHA Urban Property**: Built-up footprints, Milkat Patra property cards, and permissible Floor Space Index (FSI).
- **L6 Bank Mortgages (RBI ULI)**: Real-time institutional financial liens preventing double-financing fraud.
- **L7 RCCMS Revenue Court Cases**: Real-time litigation tracking; freezes mutation instantly upon active Stay Orders.
- **L8 Statutory Valuation**: Algorithmic calculation of circle rates eliminating valuation discretion.

Frontend Visualizer: Renders interactive SVG parcel polygons with dynamic red diagonal cross-hatching for active court disputes and golden padlock badges for bank encumbrances.

7. Cryptographic Security & Legal Compliance Architecture

- **Information Technology Act, 2000 (Section 5 & 65B)**: Electronic records digitally signed by revenue officers carry full legal presumption of authenticity in judicial proceedings.
- **Tamper-Evident SHA-256 Hashing**: When an officer approves a record, a cryptographic SHA-256 hash digest is computed across the parcel attributes, timestamp, and officer identity, embedded directly into the digitized extract and printable QR code.
- **Multi-Channel OTP Verification**: Account creation and digital signing incorporate real-time EmailJS / Resend email OTP challenges with 10-minute expiry windows and rate limiting.
- **Privacy Preservation**: Citizen Aadhaar numbers are stored exclusively in masked format (XXXX-XXXX-9124) complying with UIDAI security mandates.

8. Cloud DevOps & Deployment Infrastructure

- **Infrastructure as Code (IaC):** Declarative deployment managed via render.yaml orchestrating the Node.js Express web service (ilrd-backend) and Next.js static frontend (ilrd-frontend).
- **Automated CI/CD:** Every Git push to main triggers automated TypeScript validation (tsc), Prisma schema push (prisma db push), Next.js bundle compilation, and zero-downtime deployment.
- **Active Health Telemetry:** Backend exposes GET /health executing a live SELECT 1 query against PostgreSQL, monitoring database latency in milliseconds and guaranteeing automated self-healing restart if degraded.

9. Master Dependency & Technical Specification Matrix

Category	Package / Tool	Version	Runtime / Scope	Architectural Responsibility
Frontend Core	Next.js	14.2.24	Node.js / Browser	App Router, SSR, SSG static exports, optimized route bundling.
Frontend Core	React & React DOM	18.3.1	Browser Client	Concurrent UI rendering, hooks, state lifecycle.
State Management	@tanstack/react-query	5.64.2	Browser Client	Asynchronous server state caching, mutation invalidation, query pooling.
Styling Engine	Tailwind CSS + PostCSS	3.4.17	Build / CSS	Atomic design tokens, responsive breakpoints, government palette.
Form & Validation	React Hook Form + Zod	7.54 / 3.24	Browser Client	High-performance unopinionated form inputs with strict Zod validation.
Internationalization	i18next / react-i18next	26.4 / 17.0	Browser Client	Dynamic multi-language localization (Marathi, Hindi, English).
Data Visualization	Recharts	2.15.0	Browser Client	Interactive SVG analytics charts for revenue administrative dashboards.
QR / Barcode	qrcode / jsQR / jsbarcode	1.5.4 / 1.4 / 3.12	Browser Client	Cryptographic QR generation, camera scanning, archival barcode output.
Backend Framework	Express.js	4.21.2	Node.js Server	RESTful routing, authentication middleware, multipart streaming.
Language	TypeScript	5.7.3	Universal	End-to-end static type safety across client, server, and database models.
Database & ORM	PostgreSQL + Prisma	PG 16 / Prisma 6.4	Database Engine	ACID transactions, spatial geometry indexing, automated schema migration.
AI Vision (Cloud)	Google Gemini 1.5 Flash	v1beta Multimodal	Cloud API	Sub-2s Devanagari handwriting and tabular Satbara visual extraction.
AI Vision (Local)	OpenCV + EasyOCR	Python / On-Device	Edge Microservice	Offline edge OCR fallback: image deskewing, binarization, character parse.
Security & Auth	jsonwebtoken + bcryptjs	9.0.2 / 3.0.2	Node.js Server	Cryptographic stateless session tokens, salted password hashing.
Image Processing	Sharp + pdf-parse	0.33.5 / 1.1.1	Node.js Server	High-speed rasterization, DPI optimization, PDF text stream parsing.
Email & Notification	EmailJS Browser / API	4.4.1 / REST	Multi-Channel	Real-time email OTP verification, citizen welcome, status notifications.
Cloud DevOps	Render Cloud (IaC)	render.yaml v1	Cloud Platform	Infrastructure as Code, automated Git webhooks, zero-downtime deployment.